

AR Bouldering Replay with Contact-Aware Human-Scene Alignment

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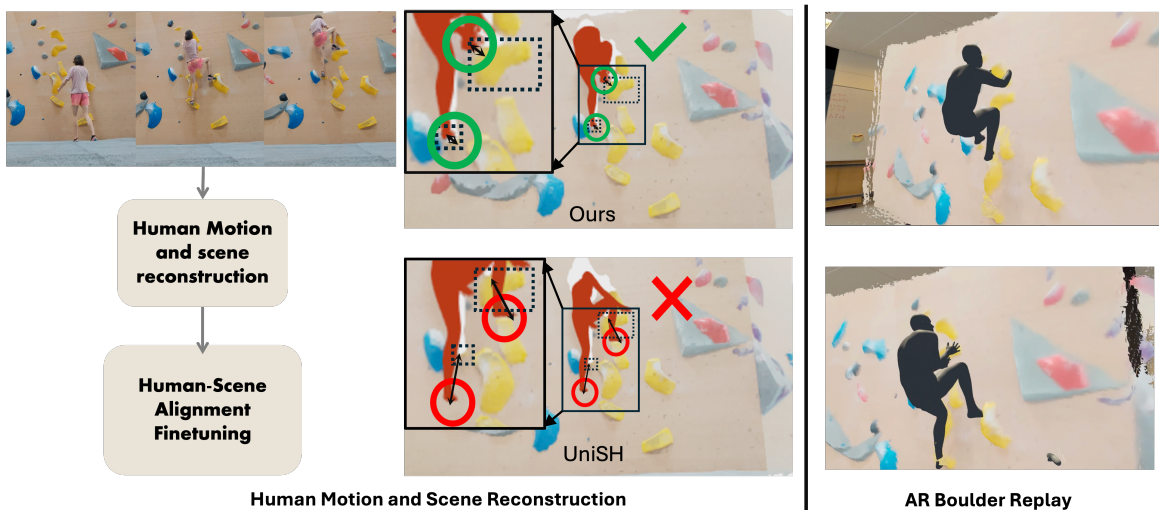


Figure 1: 4D Human-Scene Reconstruction for Bouldering: Given a monocular video, we reconstruct both the climbing wall and the climber’s motion, with the motion precisely anchored to the wall. The middle row shows the reconstruction results corresponding to the third frame of the input sequence. Circles denote the reconstructed foot positions, rectangles denote the corresponding holds, and arrows indicate the distances between each foot and its corresponding hold. Compared with the baseline, our method achieves better human–scene alignment. We further develop an AR application for immersive bouldering replay.

ABSTRACT

We present a 4D human-scene reconstruction method for precise human-scene alignment in bouldering scenarios. 4D bouldering replay can convey complex climbing movements from different perspectives and support asynchronous collaborative training. However, it is nontrivial to accurately reconstruct and align human motion with climbing-wall geometry from monocular video. To address this problem, we introduce a contact-aware fine-tuning method that builds upon state-of-the-art human-scene reconstruction methods. By incorporating human-scene contact and penetration constraints, our method improves alignment between reconstructed climbers and climbing walls. We further develop an AR bouldering system for climbing-wall visualization and motion replay in arbitrary indoor environments.

Index Terms: Human Motion Recovery, Scene Reconstruction, Boulder Training, Augmented Reality, Sports.

1 INTRODUCTION

Indoor bouldering has gained popularity as a safer alternative to outdoor climbing and a form of full-body exercise. This growth has motivated research on improving the climbing experience and supporting the climbing process. Projector-based methods project expert movements onto climbing walls from different perspectives to reduce the cognitive load of remembering climbing sequences [2].

To simplify hardware setup and provide a more immersive experience, virtual reality (VR) bouldering systems have been proposed to reduce the fear of height and falling [7] and simulate real climbing in VR [5]. However, due to the absence of tactile feedback, it remains challenging to provide a realistic climbing experience.

AR bouldering can potentially address this gap by overlaying augmented elements directly into the user’s field of view. BoulderAR proposed a phone-based AR system to record the bouldering process and visualize bouldering problems for a collaborative bouldering experience [1]. BoulderPlay gamifies the bouldering experience to provide a novel climbing experience [6]. However, existing AR bouldering systems are designed primarily for standardized climbing walls, such as MoonBoard-style walls, and may not generalize well to arbitrary climbing walls with varying surface angles and irregular hold placements. This gap mainly exists because reconstructing in-the-wild climbing walls and human motion using easily accessible devices remains challenging. Therefore, this work explores a 4D bouldering replay method from monocular videos that reconstructs both the climbing wall and human motion, with potential applications in movement reflection, training support, and asynchronous collaboration.

For joint scene and human motion reconstruction, Josh [4] shows promising results even for human movement on different terrain. However, as mentioned, when human-scene contact is invisible, the reconstruction quality degrades. UniSH [3] proposed a feed-forward joint human motion and scene reconstruction method. However, due to the domain gap in bouldering scenarios and the high requirement for precise alignment between human motion and the climbing scene, direct usage of these methods is not practical.

To address these challenges, we propose a high-accuracy scene and human motion reconstruction method that further fine-tunes the UniSH model. We use the output of the UniSH model as the initial

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reconstruction result. To further refine the human model, we introduce a projection-based human-scene contact loss and a penetration loss, which improve alignment between the human and the scene and reduce interpenetration of the human motion into the scene. For the human-scene contact loss, we implement a semi-manual annotation pipeline to label human-scene contacts. Both qualitative and quantitative evaluations show improved alignment between the scene and human motion using our method. We further develop an AR application for anchoring the reconstructed climbing wall into the arbitrary indoor room and replaying the motion.

2 METHOD

Scene and Human Motion Fine-tuning. The UniSH model outputs a scene point map and human motion in the world coordinate system, which serve as the initial reconstruction result. To further align the human motion with the scene, we introduce a contact-aware alignment loss and a penetration loss in addition to the alignment loss used during fine-tuning. For the contact-aware alignment loss, if a climber’s hands or feet are in contact with a specific hold in a given frame, we project the corresponding 3D limbs into the image space and compute the closest distance between the projected limbs and the hold mask, encouraging accurate human-scene contact. For the penetration loss, we compute both human and scene depth maps for each frame and penalize human vertices that lie behind the scene surface. Together, these losses improve human-scene alignment at contact regions while reducing geometric penetration between the reconstructed human body and the scene.

Human and Scene Contact Labeling. The contact-aware loss requires ground-truth labels indicating which climbing holds are in contact with the human limbs. To obtain these labels, we developed a two-stage semi-manual annotation pipeline. First, users click the climbing holds in the first frame, and SAM2 tracks them throughout the video to generate instance masks. Second, we annotate human-hold contacts. For selected frames, the system provides an interface that facilitates assigning the four limbs to the corresponding holds. In practice, only frames exhibiting obvious misalignment between the reconstructed human motion and the scene need to be annotated, rather than the entire video. This annotation pipeline significantly reduces labeling effort and can be generalized to other videos involving precise human-scene interactions.

4D Bouldering Replay in AR. Based on the previous steps, we obtain a reconstructed scene represented as point maps and human motion represented by the SMPL model. To export these results to Unity, several adaptations are required. First, the point maps are converted into a 3D scene mesh. Since the climber occludes different parts of the climbing wall across frames, the point maps exhibit varying levels of occlusion. To reconstruct a complete climbing wall, we align the centers of the point maps from sampled frames to a common coordinate system and accumulate them into a unified point cloud. Frames are sampled at the same interval from the video. We convert the accumulated point cloud into a mesh using the marching cubes algorithm. In this way, a complete 3D wall can be reconstructed. For human motion, a custom conversion script transforms the reconstructed motion into Unity-compatible animation clips. The reconstructed scene mesh and animation sequence are then imported into Unity. For AR replay, to simplify model anchoring, the largest vertical wall in the environment is detected automatically based on the headset’s native spatial tracking and SLAM pipeline. The floor plane of the reconstructed scene is aligned with the physical floor, and the reconstructed climbing wall is placed adjacent to the detected vertical wall.

3 EXPERIMENTS

The training and inference of the human-scene reconstruction model were conducted on a single NVIDIA RTX 5080 GPU. For 4D replay, we developed an AR application in Unity. The system

was implemented in Unity 6000.1.13f1, with the Meta Quest 3 serving as the target display platform.

To evaluate the human-scene alignment, we annotate human-scene contact information for the entire video and compute the average distance between the projected 2D human limbs and their corresponding holds. In addition, we calculate the average IoU (Intersection over Union) between the projected human mask and the ground-truth human mask. Without our fine-tuning method, the human mask IoU is 0.422, and the average limb-to-hold distance is 9.652 pixels. After fine-tuning, the mask IoU increases to 0.615, while the average distance decreases to 1.718 pixels, showing improved human motion reconstruction and human-scene alignment. As shown in Fig. 1, both feet of our avatar are correctly aligned with right holds, corresponding to the third frame of the input sequence.

4 DISCUSSION

Although we achieve promising results, several limitations remain. First, human-scene contact still requires manual annotation, even though only misaligned frames need to be labeled. A potential direction is to automate this process. Given the initial reconstruction of the human body, one could assign contact labels by searching for the nearest holds; however, this approach is sensitive to the quality of the initial reconstruction. Second, the current indoor anchoring method is suitable for pre- and post-bouldering session analysis and discussion, but it is not designed for in-situ anchoring while users are actively in front of a climbing wall. Future work could explore automatic on-site anchoring for bouldering and quantitatively evaluate registration accuracy. Finally, further user studies are needed to evaluate the effectiveness of 4D bouldering replay for self-learning and collaborative training.

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